collection of all these now-slices fills out a substantial region of the spacetime loaf. In fact, if space is infinite—if now-slices extended infinitely far—then the rotated now-slices can be centered arbitrarily far away, and hence their union sweeps through *every* point in the spacetime loaf.[9]

So: *if you buy the notion that reality consists of the things in your freeze-frame mental image right now, and if you agree that your* now *is no more valid than the* now *of someone located far away in space who can move freely, then reality encompasses all of the events in spacetime.* The total loaf exists. Just as we envision all of space as *really* being out there, as *really* existing, we should also envision all of time as *really* being out there, as *really* existing, too. Past, present, and future certainly appear to be distinct entities. But, as Einstein once said, "For we convinced physicists, the distinction between past, present, and future is only an illusion, however persistent."[5] The only thing that's real is the whole of spacetime.

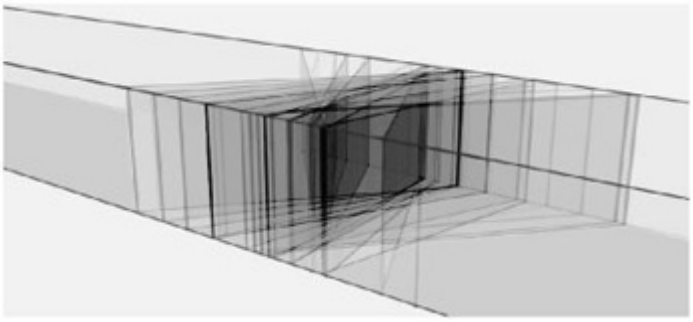


Figure 5.5 A sample of now-slices for a variety of observers (real or hypothetical) situated at a variety of distances from earth, moving with a variety of velocities.

Experience and the Flow of Time

In this way of thinking, events, regardless of when they happen from any particular perspective, just *are.* They all exist. They eternally occupy their particular point in spacetime. There is no flow. If you were having a great time at the stroke of midnight on New Year's Eve, 1999, you still are, since that is just one immutable location in spacetime. It is tough to accept this description, since our worldview so forcefully distinguishes